Receiver Operating Characteristic (ROC)

1. **Question:** What is the purpose of the **Receiver Operating Characteristic (ROC)** curve?
   Answer: To evaluate the performance of a binary classifier across different threshold values.
   Explanation: The ROC curve shows how sensitivity (TPR) and specificity (1−FPR) change as the classification threshold varies.

2. **Question:** What two rates are plotted against each other in the ROC curve?
   Answer: True Positive Rate (TPR) vs False Positive Rate (FPR).
   Explanation: TPR is plotted on the Y-axis, and FPR is on the X-axis.

3. **Question:** What is the formula for the **True Positive Rate (TPR)**?
   Answer: ($\text{TPR} = \frac{TP}{TP + FN}$)
   Explanation: Measures the proportion of actual positives correctly identified.

4. **Question:** What is the formula for the **False Positive Rate (FPR)**?
   Answer: ($\text{FPR} = \frac{FP}{FP + TN}$)
   Explanation: Measures the proportion of negatives incorrectly classified as positives.

5. **Question:** What probability value is varied to trace the ROC curve?
   Answer: The classification threshold probability (p^*).
   Explanation: Changing the threshold affects TPR and FPR values, generating the ROC curve.

6. **Question:** What two other metrics can be used to understand binary classifier performance?
   Answer: Precision and Recall.
   Explanation: These are threshold-dependent metrics, unlike AUC which is threshold-independent.

7. **Question:** When ($p^* = 1$), what is the resulting **True Positive Rate (TPR)**?
   Answer: 0
   Explanation: No samples are predicted as positive, so TPR = 0.

8. **Question:** When ($p^* = 1$), what is the resulting **False Positive Rate (FPR)**?
   Answer: 0
   Explanation: No positives are predicted, so there are no false positives.

9. **Question:** When ($p^* = 0$), what is the resulting **True Positive Rate (TPR)**?
   Answer: 1

Explanation: All samples are classified positive, so all true positives are captured.

10. **Question:** What are the TPR and FPR values for a **perfect model** on the ROC curve?
    Answer: TPR = 1, FPR = 0
    Explanation: The model perfectly distinguishes positives and negatives.

11. **Question:** What does the ROC curve of a **completely random model** look like?
    Answer: A diagonal line from (0,0) to (1,1).
    Explanation: Equal probability of true and false positives—no discrimination.

12. **Question:** For a real model, what can the ROC curve help choose?
    Answer: The optimal classification threshold.
    Explanation: Based on the tradeoff between sensitivity and specificity.

Area Under the Curve (AUC)

13. **Question:** What does **AUC** stand for?
    Answer: Area Under the Curve.
    Explanation: Refers to the area under the ROC curve.

14. **Question:** AUC-ROC is a **threshold-independent** metric to assess what?
    Answer: The overall discriminative ability of the classifier.
    Explanation: It measures how well the model distinguishes between classes.

15. **Question:** What is the **AUC-ROC** value for a perfect model?
    Answer: 1.0
    Explanation: A perfect model achieves full separation of positive and negative classes.

16. **Question:** What is the **AUC-ROC** value for a completely random model?
    Answer: 0.5
    Explanation: Random guessing leads to a diagonal ROC with area 0.5.

17. **Question:** What is the typical range of AUC-ROC for a real model?
    Answer: Between 0.5 and 1.0
    Explanation: Higher values mean better classification performance.

Multi-Class Classification & Softmax

18. **Question:** What classification problem involves more than two classes?
    Answer: Multi-class classification.
    Explanation: The model must predict among three or more categories.

19. **Question:** What two names are given for the regression technique used for multi-class classification?
    Answer: Softmax Regression or Multinomial Logistic Regression.
    Explanation: Both refer to the same algorithm.

20. **Question:** What mathematical function evaluates class probabilities ($P(y=k|x)$)?
    Answer: The Softmax function.
    Explanation: Converts linear outputs into normalized probabilities.

21. **Question:** What does the index (k) denote in softmax probability?
    Answer: The specific class label.
    Explanation: Each (k) corresponds to a distinct class in the output.

22. **Question:** What does (K) represent in the categorical cross-entropy loss function?
    Answer: The total number of classes.
    Explanation: Summation in the loss function runs over all classes.

23. **Question:** How is the final predicted value ($\hat{y}$) chosen in Softmax Regression?
    Answer: By selecting the class with the highest probability.
    Explanation: ($\hat{y} = \text{argmax}_k P(y=k|x)$)

24. **Question:** What is the name of the loss function used in Softmax Regression?
    Answer: Categorical Cross-Entropy Loss.
    Explanation: Measures the difference between predicted and true class distributions.

25. **Question:** What must the range be for the evaluated probability ($P(y=k|x)$)?
    Answer: Between 0 and 1.
    Explanation: Softmax ensures all probabilities are positive and sum to 1.

Application Examples

26. **Question:** Application in **Mechanical Engineering**?
    Answer: Fault diagnosis in rotating machinery.
    Explanation: Classification used for predictive maintenance.

27. **Question:** Application in **Civil Engineering**?
    Answer: Crack detection in concrete structures.
    Explanation: Image-based classification used for structural safety.

28. **Question:** Application in **Aerospace Engineering**?
    Answer: Aircraft fault detection or flight anomaly detection.
    Explanation: Ensures safety through pattern recognition.

29. **Question:** Application in **Chemical Engineering**?
    Answer: Process fault classification in chemical plants.
    Explanation: Detects abnormal reactions or equipment failure.

30. **Question:** Application in **Environmental Engineering**?
    Answer: Air or water quality classification.
    Explanation: Models used to classify pollution levels from sensor data.